Magneto-optical conductivity of double Weyl semimetals

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We investigate the magneto-optical response of double Weyl semimetals whose energy dispersion is intrinsically anisotropic. We find that in the presence of a magnetic field, the most salient feature of the optical conductivity is a series of resonant peaks with the corresponding frequencies scaling linearly with the strength of the magnetic field. In addition, the optical conductivity is found to be anisotropic, with two of the three longitudinal components residing at a linear background and the remaining one at a constant background. The effects of chemical potential, temperature, impurity scattering, and particle-hole symmetry breaking on the optical conductivity are also studied.

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I. INTRODUCTION

Recently, topological semimetals have been actively studied because of the existence of nontrivial band crossings protected by topology and symmetries [1]. According to the dimension and degeneracy of the band crossings, the currently most studied topological semimetals can be classified into three classes: Weyl semimetals [2–9], Dirac semimetals [10–16], and nodal line semimetals [17–21]. For the Weyl semimetals, the band crossings are isolated points (zero dimension) with twofold degeneracy, known as Weyl points, and play the role of monopoles in the Brillouin zone. As the net charge of the monopoles in the Brillouin zone must be zero, the number of Weyl points with opposite monopole charge must be equal [22].

The monopole charge of a Weyl point is equal to the number of Berry fluxes passing through a closed surface enclosing only the target Weyl point. So far, most studies on Weyl semimetals have focused on the case with the lowest monopole charge, $C = \pm 1$ (we will refer to Weyl semimetals with $C = \pm 1$ as single Weyl semimetals for notational simplicity) [23–27]. The reason that this case has attracted special research interest is because the energy dispersion away from the Weyl points is linear in all directions and thus an analog of the elementary Weyl fermions in particle physics; furthermore, it is the one that has been found in condensed-matter experiments to date [26]. However, the much more unexplored Weyl points with higher monopole charge [28–41] are also of fundamental interest in at least two aspects. First, the increase in monopole charge no doubt will enhance a series of effects predicted or observed in single Weyl semimetals, like the anomalous Hall effect [42–45] and the ones related to chiral anomaly [46–53]. Second, the energy dispersion will become intrinsically anisotropic for higher monopole charge, which consequently will induce novel physics absent in single Weyl semimetals. For instance, it is found that Weyl points with higher monopole charge exhibit anisotropic screening of the Coulomb interactions [31–34].

In this work, we will investigate the magneto-optical conductivity of Weyl semimetals with $C = \pm 2$, the so-called double Weyl semimetals, of which $HgCr_2Se_4$ [28,29] and $SrSi_2$ [30] were predicted as candidate materials. In the presence of a magnetic field, the continuum spectrum of the Weyl Hamiltonian will transform to a series of discrete Landau levels whose extrema have a divergent density of states. When an optical transition between two Landau levels takes place at their extrema, a resonant peak will show up in the real part of the optical conductivity. In experiments, from these resonant peaks, information like the energy gap and the Fermi velocity of the underlying band structure can be extracted [54–60]. For instance, it is known that when the frequencies of the resonant peaks ω and the magnetic field B follow an $\omega \propto \sqrt{B}$ law, the energy dispersion of the Hamiltonian is gapless and linear, like those in the Dirac semimetals and single Weyl semimetals [61–63]. For the double Weyl semimetals concerned in this paper, we find the resonant peaks of the optical conductivity exhibit several distinctive features which can thus be applied to determine whether a material is a double Weyl semimetal or not in experiments.

This paper is organized as follows. In Sec. II, we give the Hamiltonian and the formula for the optical conductivity. In Sec. III, the numerical results of the conductivity are given. In Sec. IV, we give our estimation of the conditions needed for the magneto-optical experiments. A discussion and conclusions are presented in Sec. V.

II. FORMALISM

As Weyl points are crossings of two bands without degeneracy, the low-energy effective Hamiltonian of double Weyl semimetals is given by ($\hbar = c = k_B = 1$)

$$H_\chi(\mathbf{k}) = \lambda\left(k_x^2 - k_y^2\right)\tau_x + 2\lambda k_x k_y \tau_y + \chi v k_z \tau_z, \qquad (1)$$

where τ_i are Pauli matrices, λ is a constant parameter with the dimension of inverse mass, v refers to the velocity in the direction of linear dispersion, and $k_{x,y,z}$ refer to the momenta relative to the Weyl points; $\chi = \pm 1$ denote two kinds of chirality. Without loss of generality, v and λ are assumed to be positive. The energy spectra can be readily obtained,

$$E_{\pm,\chi}(\mathbf{k}) = \pm\sqrt{v^2 k_z^2 + \lambda^2\left(k_x^2 + k_y^2\right)^2}, \qquad (2)$$

which are linear along the z direction and quadratic in the x-y plane.

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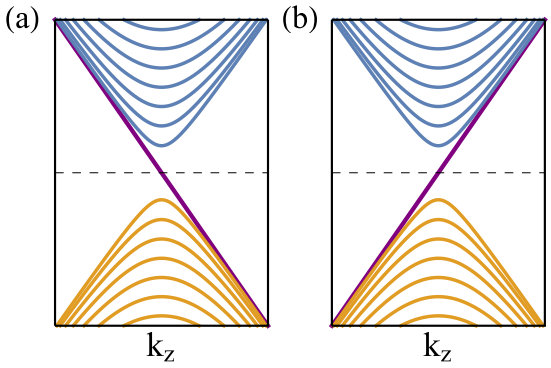


FIG. 1. The Landau levels of double Weyl semimetals. (a) $\chi = 1$; (b) $\chi = -1$. The Landau levels in blue (yellow) belongs to the particle (hole) branch. For the particle (hole) branch, the Landau level index n_p (n_h) is 2,3, ... from bottom (top) to top (bottom). In both panels, the chiral Landau levels ($n = 0,1$) are in purple and doubly degenerate. The dashed line represents the Fermi level.

Now we consider a magnetic field that is exerted along the z direction, $\mathbf{B} = B\hat{z}$. Following the standard minimal coupling, i.e., $\mathbf{k} \to \Pi = \mathbf{k} + e\mathbf{A}$, with $A = (0, Bx, 0)$, the Hamiltonian in Eq. (1) is rewritten as

$$H_\chi(\Pi) = \lambda\left(\Pi_x^2 - \Pi_y^2\right)\tau_x + \lambda(\Pi_x\Pi_y + \Pi_y\Pi_x)\tau_y + \chi v\Pi_z. \tag{3}$$

Note that Π_x and Π_y do not commute, so we have adopted the symmetric ordering for operators. The magnetic field will discretize the continuum energy spectra into a series of Landau levels (or Landau bands), whose form can be easily obtained by introducing the following ladder operators:

$$a = \frac{l_B}{\sqrt{2}}(\Pi_x - i\Pi_y), \quad a^\dagger = \frac{l_B}{\sqrt{2}}(\Pi_x + i\Pi_y), \tag{4}$$

where $l_B = 1/\sqrt{eB}$ is the magnetic length. Inserting Eq. (4) into Eq. (3) and rewriting the Hamiltonian into a matrix form,

$$H = \begin{pmatrix} \chi v k_z & \frac{2\lambda}{l_B^2}a^2 \\ \frac{2\lambda}{l_B^2}a^{\dagger 2} & -\chi v k_z \end{pmatrix}, \tag{5}$$

it is readily found that the Landau levels are given by

$$E_{n,s,\chi}(k_z) = \begin{cases} sE_n, & n \geqslant 2, \\ -\chi v k_z, & n = 0,1, \end{cases} \tag{6}$$

where $E_n(k_z) \equiv \sqrt{v^2k_z^2 + 4n(n-1)(\lambda eB)^2}$; $s = +$ and $s = -$ correspond to the particle branch and hole branch, respectively, and when $n = 0$ or 1, $s = +$ is implicitly assumed. A graph illustration of the Landau levels is shown in Fig. 1. Before proceeding, we make two remarks on the Landau levels: (i) $E_n(0) \propto B$, and such a linear dependence is distinct from the $E_n(0) \propto \sqrt{B}$ law found in the Dirac and single Weyl semimetals [61,62]; (ii) there are two degenerate chiral Landau levels ($n = 0,1$), with the number equating to the monopole charge of the Weyl points. As we will show in Sec. III D, when the degeneracy is lifted, an optical transition between the two chiral Landau levels will take place.

For $n \geqslant 2$, we denote the eigenvectors corresponding to $E_{n,s,\chi}(k_z)$ as $|n,s,\chi\rangle = (\alpha_{n,s,\chi}|n-2\rangle, \beta_{n,s,\chi}|n\rangle)^T$, where $|n\rangle$ satisfies $a|n\rangle = \sqrt{n}|n-1\rangle$ and $a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$. The coefficients $\alpha_{n,s,\chi}$ and $\beta_{n,s,\chi}$ can be determined straightforwardly, with

$$\alpha_{n,s,\chi} = s\sqrt{\frac{E_{n,s,\chi} + \chi v k_z}{2E_{n,s,\chi}}}, \quad \beta_{n,s,\chi} = \sqrt{\frac{E_{n,s,\chi} - \chi v k_z}{2E_{n,s,\chi}}}. \tag{7}$$

For $n = 0,1$, the eigenvectors are simply $(0,1)^T$.

The optical conductivity can be obtained from the Kubo formula, which is

$$\sigma_{\mu\nu}(\omega) = \frac{-i}{2\pi l_B^2}\sum_{n,n',s,s',\chi}\int\frac{dk_z}{2\pi}\frac{f_{n,s,\chi} - f_{n',s',\chi}}{E_{n,s,\chi} - E_{n',s',\chi}} \times \frac{\langle n,s,\chi|j_{\mu,\chi}|n',s',\chi\rangle\langle n',s',\chi|j_{\nu,\chi}|n,s,\chi\rangle}{\omega - E_{n,s,\chi} + E_{n',s',\chi} + i\Gamma}, \tag{8}$$

where $f_{n,s,\chi} = 1/[\exp(E_{n,s,\chi} - \mu)/T + 1]$ is the Fermi-Dirac distribution function, with μ being the chemical potential and T being the temperature; Γ denotes the impurity scattering rate (in this work, we assume that all Landau levels share the same Γ for simplicity); and $j_{\mu,\chi}$ ($\mu = x,y,z$) denote the current operators, whose explicit forms are

$$j_{x,\chi} = \frac{\partial H_\chi}{\partial A_x} = 2e\lambda[\Pi_x\sigma_x + \Pi_y\sigma_y] = 2\sqrt{2}\frac{e\lambda}{l_B}\begin{pmatrix} 0 & a \\ a^\dagger & 0 \end{pmatrix},$$

$$j_{y,\chi} = \frac{\partial H_\chi}{\partial A_y} = 2e\lambda[-\Pi_y\sigma_x + \Pi_x\sigma_y] = i2\sqrt{2}\frac{e\lambda}{l_B}\begin{pmatrix} 0 & -a \\ a^\dagger & 0 \end{pmatrix},$$

$$j_{z,\chi} = \frac{\partial H_\chi}{\partial A_z} = \chi ev\sigma_z = \chi ev\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{9}$$

In the clean limit, i.e., $\Gamma = 0$, the dissipative components which correspond to the absorption of light are found to be

$$\mathrm{Re}[\sigma_{xx}(\omega)] = -\frac{4e^2\lambda^2}{l_B^4}\sum_{n=1}n\int\frac{dk_z}{2\pi}\left[\frac{f_{n,+} - f_{n,-} - f_{n+1,+} + f_{n+1,-}}{E_n - E_{n+1}}\left(1 - \frac{v^2k_z^2}{E_nE_{n+1}}\right)\delta(\omega + E_n - E_{n+1}) + \frac{f_{n,+} - f_{n,-} + f_{n+1,+} - f_{n+1,-}}{E_n + E_{n+1}}\left(1 + \frac{v^2k_z^2}{E_nE_{n+1}}\right)\delta(\omega - E_n - E_{n+1})\right],$$

$$\mathrm{Re}[\sigma_{zz}(\omega)] = -\frac{e^2v^2}{l_B^2}\sum_{n=2}\int\frac{dk_z}{2\pi}\frac{f_{n,+} - f_{n,-}}{2E_n}\left(1 - \frac{v^2k_z^2}{E_n^2}\right)\delta(\omega - 2E_n),$$

$$\mathrm{Im}[\sigma_{\mathrm{xy}}(\omega)] = -\frac{4e^2\lambda^2}{l_B^4}\sum_{n=1} n\int\frac{dk_z}{2\pi}\Bigg[\frac{-f_{n,+}-f_{n,-}+f_{n+1,+}+f_{n+1,-}}{E_n-E_{n+1}}\left(1-\frac{v^2k_z^2}{E_nE_{n+1}}\right)\delta(\omega+E_n-E_{n+1}) + \frac{f_{n,+}+f_{n,-}-f_{n+1,+}-f_{n+1,-}}{E_n+E_{n+1}}\left(1+\frac{v^2k_z^2}{E_nE_{n+1}}\right)\delta(\omega-E_n-E_{n+1})\Bigg], \tag{10}$$

where we have defined $f_{n,\pm}\equiv 1/[\exp(\pm E_n-\mu)/T+1]$ for simplicity of notation.

Note that $\mathrm{Re}(\sigma_{\mathrm{yy}})=\mathrm{Re}(\sigma_{\mathrm{xx}})$ due to rotational symmetry. From Eqs. (8) and (9), it is readily found that the other two transverse components σ_{xz} and σ_{yz} are zero. The selection rules of the optical transition can be immediately read out from the δ functions in Eq. (10). The δ functions in the first line of $\mathrm{Re}(\sigma_{\mathrm{xx}})$ and $\mathrm{Im}(\sigma_{\mathrm{xy}})$ denote the transitions between nearest-neighbor Landau levels within the same branch; that is, the selection rules are $\Delta n=n_{p,f}-n_{p,i}=1$ and $\Delta n=n_{h,f}-n_{h,i}=-1$, where $n_{p(h),i}$ and $n_{p(h),f}$ denote the indices of the initial particle (hole) Landau level and the final particle (hole) Landau level of the transition process, respectively (see the caption of Fig. 1). The δ functions in the second line of $\mathrm{Re}(\sigma_{\mathrm{xx}})$ and $\mathrm{Im}(\sigma_{\mathrm{xy}})$ indicate that the transitions can also occur between the particle-branch Landau levels and the hole-branch Landau levels, with the selection rules $\Delta n=n_{p,f}-n_{h,i}=\pm1$. The δ function in $\mathrm{Re}(\sigma_{zz})$ indicates that the transitions can occur only between the particle-branch Landau levels and the hole-branch Landau levels, with the selection rule $\Delta n=n_{p,f}-n_{h,i}=0$.

After performing the integration over k_z, we find that

$$\mathrm{Re}[\sigma_{\mathrm{xx}}(\bar\omega)] = \frac{2e^2\lambda}{\pi v l_B^2}\sum_{n=1}^{\xi(\bar\omega)}\left[-\frac{\sinh\left(\frac{2n-\bar\omega^2}{2\bar T\bar\omega}\right)}{\cosh\left(\frac{2n-\bar\omega^2}{2\bar T\bar\omega}\right)+\cosh\left(\frac{\bar\mu}{\bar T}\right)}+\frac{\sinh\left(\frac{2n+\bar\omega^2}{2\bar T\bar\omega}\right)}{\cosh\left(\frac{2n+\bar\omega^2}{2\bar T\bar\omega}\right)+\cosh\left(\frac{\bar\mu}{\bar T}\right)}\right]\frac{n}{\bar\omega}\frac{|2n^2-\bar\omega^2|}{\sqrt{4n^2-4n^2\bar\omega^2+\bar\omega^4}}\theta(|\sqrt{2n}-\bar\omega|),$$

$$\mathrm{Re}[\sigma_{zz}(\bar\omega)] = \frac{e^2v}{2\pi\lambda}\frac{\sinh\left(\frac{\bar\omega}{2\bar T}\right)}{\cosh\left(\frac{\bar\mu}{\bar T}\right)+\cosh\left(\frac{\bar\omega}{2\bar T}\right)}\sum_{n=2}^{\lfloor\frac{1+\sqrt{\bar\omega^2+1}}{2}\rfloor}\frac{n^2-n}{\bar\omega^2\sqrt{4n-4n^2+\bar\omega^2}},$$

$$\mathrm{Im}[\sigma_{\mathrm{xy}}(\bar\omega)] = -\frac{2e^2\lambda}{\pi v l_B^2}\sum_{n=1}^{\xi(\bar\omega)}\left[\frac{\cosh\left(\frac{2n-\bar\omega^2}{2\bar T\bar\omega}\right)+\exp\left(\frac{\bar\mu}{\bar T}\right)}{\cosh\left(\frac{2n-\bar\omega^2}{2\bar T\bar\omega}\right)+\cosh\left(\frac{\bar\mu}{\bar T}\right)}-\frac{\cosh\left(\frac{2n+\bar\omega^2}{2\bar T\bar\omega}\right)+\exp\left(\frac{\bar\mu}{\bar T}\right)}{\cosh\left(\frac{2n+\bar\omega^2}{2\bar T\bar\omega}\right)+\cosh\left(\frac{\bar\mu}{\bar T}\right)}\right]\frac{n}{\bar\omega}\frac{|2n^2-\bar\omega^2|}{\sqrt{4n^2-4n^2\bar\omega^2+\bar\omega^4}}\theta(|\sqrt{2n}-\bar\omega|), \tag{11}$$

where $\bar\omega\equiv\omega l_B^2/2\lambda$, $\bar\mu\equiv\mu l_B^2/2\lambda$, and $\bar T\equiv Tl_B^2/2\lambda$ are dimensionless quantities and the upper bound of the summation over n, $\xi(\bar\omega)$, is given by

$$\xi(\bar\omega)=\begin{cases}\left\lfloor\frac{\bar\omega^2}{2\sqrt{\bar\omega^2-1}}\right\rfloor, & \text{if } \bar\omega>1,\\ \infty, & \text{otherwise.}\end{cases} \tag{12}$$

The expressions of the optical conductivity are too complicated to obtain analytical results. Thus, we will resort to numerical calculations in the following discussions.

III. MAGNETO-OPTICAL CONDUCTIVITY

A. Longitudinal component

Figure 2 shows the real part of the longitudinal conductivity at the neutrality condition, i.e., $\mu=0$. For both $\mathrm{Re}(\sigma_{xx})$ and $\mathrm{Re}(\sigma_{zz})$, the most prominent feature is a series of resonant peaks originating from the optical transitions between Landau levels. Due to the difference in selection rules, it is found that the resonant peaks for $\mathrm{Re}(\sigma_{xx})$ are located at $\omega=2[\sqrt{n(n+1)}+\sqrt{n(n-1)}]\lambda eB$, while for $\mathrm{Re}(\sigma_{zz})$, they are located at $\omega=4\sqrt{n(n-1)}\lambda eB$. It is immediately seen that the frequencies of the resonant peaks for both $\mathrm{Re}(\sigma_{xx})$ and $\mathrm{Re}(\sigma_{zz})$ follow an $\omega\propto B$ law, which is distinct from the $\omega\propto\sqrt{B}$ law obeyed by the Dirac semimetals and single Weyl semimetals [61,62]. In experiments, the frequency ratio between neighboring peaks is an important quantity for figuring out the Landau level indices related to the peaks and further extracting information about the underlying band structure, e.g., whether an energy gap exists or not, and the value of Fermi velocity [54,55,58–60]. Here the frequency ratio between the $(n+1)$th and nth (counting from the left) resonant peaks for $\mathrm{Re}(\sigma_{xx})$ is given by $[\sqrt{(n+2)(n+1)}+\sqrt{n(n+1)}]/[\sqrt{n(n+1)}+\sqrt{n(n-1)}]$, and for $\mathrm{Re}(\sigma_{zz})$ it is $\sqrt{n+2}/\sqrt{n}$. Furthermore, the energy offset between the $(n+1)$th and nth peaks is found to decrease with the increase in n and saturates to $4\lambda eB$ for large n; such a behavior is also distinct from the Dirac and single Weyl semimetals, in which the energy offset will monotonically decrease to zero [61,62].

Besides the difference in the locations of the resonant peaks, another distinctive feature between $\mathrm{Re}(\sigma_{xx})$ and $\mathrm{Re}(\sigma_{zz})$ is that the resonant peaks of the former sit on a linear background, while those of the latter sit on a constant background (see insets of Fig. 2). Both observations agree with the fact that without a magnetic field, the optical conductivity $\mathrm{Re}(\sigma_{xx})$ is linear in frequency, whereas $\mathrm{Re}(\sigma_{zz})$ is a constant independent of frequency [37]. Such a distinction originates from the apparent discrepancy of dispersion relations between the x-y plane and z direction and is not present in single Weyl semimetals, for which all longitudinal components should reside in a linear background [62].

Figure 3 further illustrates the effect of the chemical potential μ on the longitudinal conductivity. For $\mathrm{Re}(\sigma_{xx})$, when μ is tuned away from the neutrality condition but still crosses only the chiral Landau levels, i.e., $\bar\mu<\sqrt{2}$ (due to particle-hole symmetry, we will consider only the $\mu>0$ case), the relatively weak peak at $\bar\omega=\sqrt{2}$ shown in

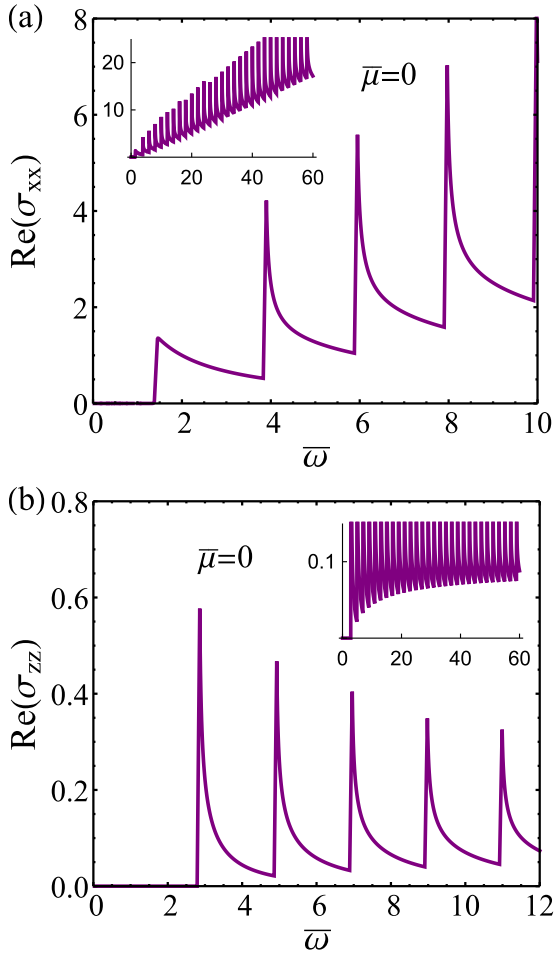


FIG. 2. The real part of the longitudinal conductivity at $\mu = 0$. Here $Tl_B^2/2\lambda = 0.01, \Gamma = 0$. (a) For Re($\sigma_{xx}$) (in units of $2e^2\lambda/\pi v l_B^2$), a series of resonant peaks is observed at $\bar{\omega} = \sqrt{n(n+1)} + \sqrt{n(n-1)}$. (b) For Re($\sigma_{zz}$) (in units of $e^2 v/2\pi\lambda$), the peaks are located at $\bar{\omega} = 2\sqrt{n(n-1)}$. A broader frequency range of the optical conductivity is shown in the insets, from which it is readily seen that for Re(σ_{xx}), the minima of the optical conductivity exhibit a linear dependence on the frequency, while for Re(σ_{zz}), the minima of the optical conductivity approach a constant for larger frequency.

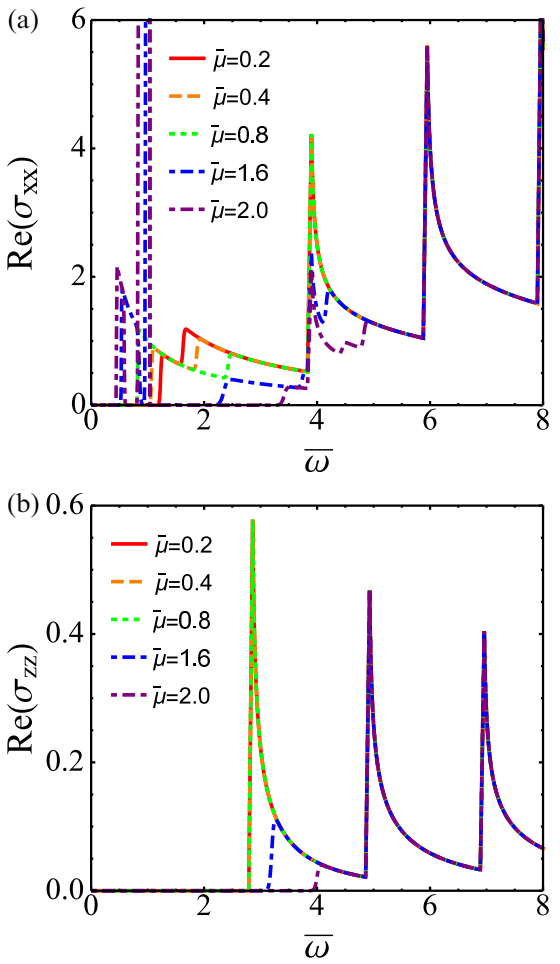


FIG. 3. The effect of the chemical potential on the real part of the longitudinal conductivity. Here $Tl_B^2/2\lambda = 0.01$, $\Gamma = 0$. (a) For Re(σ_{xx}) (in units of $2e^2\lambda/\pi v l_B^2$), the absorption edges related to the chiral Landau levels are moving with the variation of μ. When μ crosses the minimum of a Landau level, a new resonant peak shows up. (b) For Re(σ_{zz}) (in units of $e^2 v/2\pi\lambda$), when μ crosses the minimum of a Landau level, the resonant peak related to the Landau level changes to an absorption edge whose position is directly determined by μ.

Fig. 2(a) will be split into two absorption edges which are located at $\bar{\omega}_1 = \sqrt{\bar{\mu}^2+2} - \bar{\mu}$ and $\bar{\omega}_2 = \sqrt{\bar{\mu}^2+2} + \bar{\mu}$, as shown in Fig. 3(a). For Re(σ_{zz}), when μ crosses only the chiral Landau levels, however, the change in μ has no effect on the optical conductivity; this is because the selection rule for Re(σ_{zz}) is $\Delta n = n_{p,f} - n_{h,i} = 0$, with $n_{p(h)} \geqslant 2$. When μ is tuned to cross the nth particle Landau level with $n \geqslant 2$, we find that for Re(σ_{xx}), a new resonant peak will appear at $\bar{\omega} = \sqrt{n(n+1)} - \sqrt{n(n-1)}$ [the case with $n = 2$ is shown in Fig. 3(a)]. The new peak corresponds to the intrabranch transition between the nth and $(n+1)$th particle Landau levels. The transition is now allowed because the nth particle Landau level gets occupied while the $(n+1)$th particle Landau level remains empty. For Re(σ_{zz}), when μ is tuned to cross the nth particle Landau level, the resonant peak at $\bar{\omega} = 2\sqrt{n(n-1)}$ will disappear, and an absorption edge will appear at $\bar{\omega} = 2\bar{\mu}$ [the case with $n = 2$ is shown in Fig. 3(b)].

B. Transverse component

The transverse (Hall) conductivity Im(σ_{xy}) at $\mu = 0$ vanishes for all frequencies; this fact can be read directly from the expression of Im(σ_{xy}) in Eq. (11). A more physical explanation is that the transverse conductivity has four contributions: (I) the transition from the nth hole-branch Landau level to the $(n+1)$th particle-branch Landau level, (II) the transition from the $(n+1)$th hole-branch Landau level to the nth particle-branch Landau level, (III) the transition from the $(n+1)$th hole-branch Landau level to the nth hole-branch Landau level, and (IV) the transition from the nth particle-branch Landau level to the $(n+1)$th particle-branch Landau level. Importantly, the contributions from I and III have a sign difference from the ones from II and IV, respectively. Moreover, at $\mu = 0$, due to particle-hole symmetry the contributions exactly cancel each other out; thus, Im(σ_{xy}) vanishes for all frequencies in this case. Away from the neutrality condition, particle-hole symmetry is broken, and consequently, the transverse conductivity Im(σ_{xy})

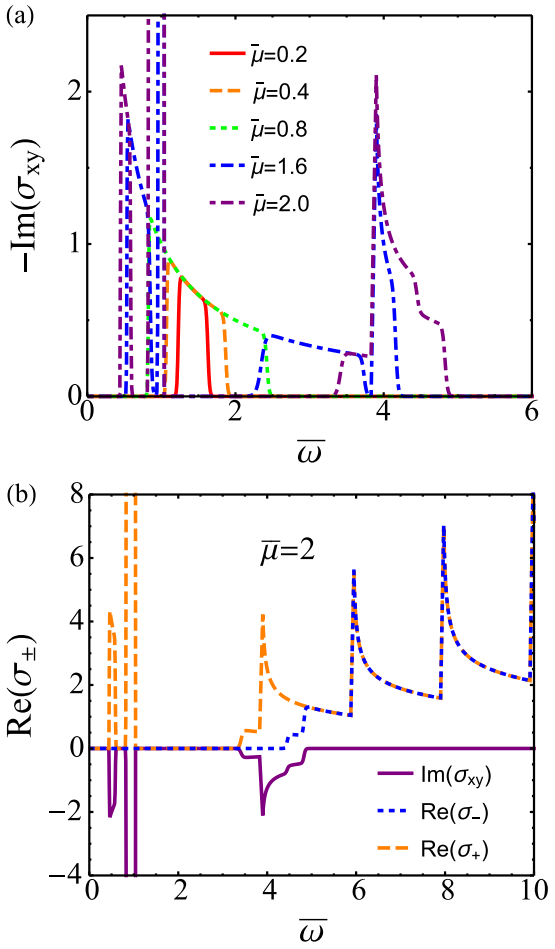


FIG. 4. The absorption part of the transverse conductivity (in units of $2e^2\lambda/\pi v l_B^2$). Here $Tl_B^2/2\lambda = 0.01$, $\Gamma = 0$. (a) Im(σ_{xy}) for several values of μ. (b) The absorption part of the optical conductivity for circularly polarized light.

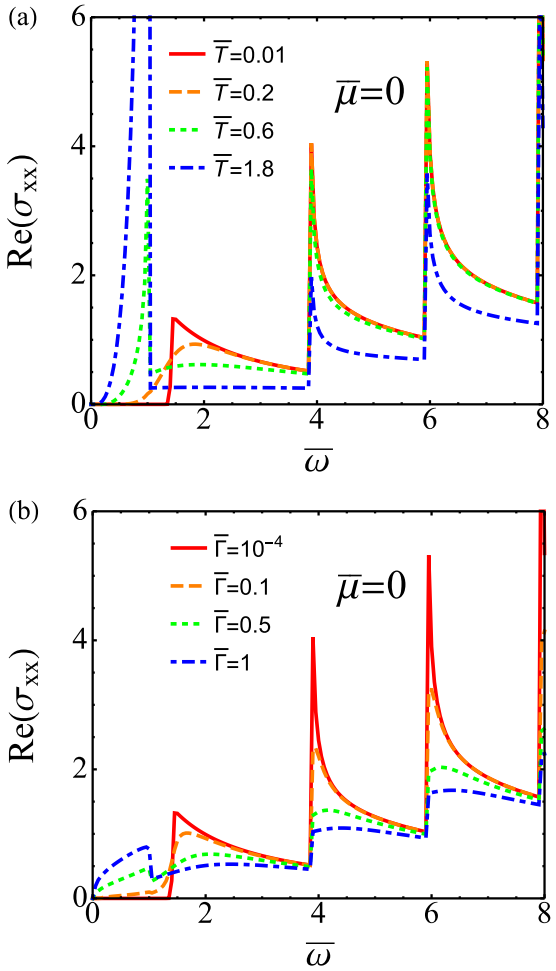


FIG. 5. The effect of temperature and impurity scattering on the optical conductivity (in units of $2e^2\lambda/\pi v l_B^2$); (a) is plotted in $\Gamma l_B^2/2\lambda = 10^{-4}$, and (b) is plotted in $Tl_B^2/2\lambda = 0.01$. A suppression of resonant peaks is observed in both cases.

will take nonzero values in certain frequency regions, as shown in Fig. 4(a).

In experiments, the quantities $\sigma_\pm \equiv \sigma_{xx} \pm i\sigma_{xy}$ are also of interest because they can be used to determine the polarization of lights. Concretely, σ_+ corresponds to right-handed polarized light, and σ_- corresponds to left-handed polarized light. The absorption parts of $\sigma_\pm$ are shown in Fig. 4(b). In the lower-frequency regime, i.e., $\omega \lesssim 2\mu$, Re(σ_-) vanishes as the longitudinal conductivity cancels the transverse conductivity. The picture is similar to that of the single Weyl semimetals [62].

C. Temperature effect and impurity scattering effect

The effect of finite temperature on the optical conductivity enters through the Fermi-Dirac distribution function. Figure 5(a) shows the temperature effect on Re(σ_{xx}) at $\mu = 0$; it is readily seen that with the increase in temperature, the height of the resonant peaks will be suppressed, and resonant peaks corresponding to intrabranch transition which are absent at zero temperature will show up (see the blue dashed line and the green dashed line). The effect of impurity scattering on Re(σ_{xx}) is shown in Fig. 5(b). It is readily seen that the main effect of the impurity scattering is just a suppression of the resonant peaks.

For Re(σ_{zz}) under the neutrality condition, finite temperature and impurity scattering mainly induce a suppression of the resonant peaks; no new peaks will show up (not shown here).

D. Effect of particle-hole symmetry breaking

Next, we will explore the effect of particle-hole symmetry breaking since in real materials there is no constraint to preserve this symmetry. To achieve this, we add a diagonal quadratic term of the form $\frac{k_x^2+k_y^2}{2m}\tau_0$ to the original Hamiltonian in Eq. (1), with $m \gtrsim \lambda^{-1}$. The Hamiltonian with standard minimum coupling in matrix form thus changes to

$$H = \begin{pmatrix} \chi v k_z + sb & \frac{2\lambda}{l_B^2}a^2 \\ \frac{2\lambda}{l_B^2}a^{\dagger 2} & -\chi v k_z + sb \end{pmatrix}, \tag{13}$$

with the symmetry-breaking term $sb = \omega_c(a^\dagger a + \frac{1}{2})$, where $\omega_c = eB/m$. Correspondingly, the Landau levels

become

$$E_{n,s,\chi}(k_z) = s\sqrt{(\chi v k_z - \omega_c)^2 + 4n(n-1)(\lambda eB)^2} + \left(n - \frac{1}{2}\right)\omega_c, \quad n \geqslant 2,$$
$$E_{1,s,\chi}(k_z) = -\chi v k_z + \frac{3}{2}\omega_c,$$
$$E_{0,s,\chi}(k_z) = -\chi v k_z + \frac{1}{2}\omega_c. \tag{14}$$

As the symmetry-breaking term does not change the selection rules (this can be simply confirmed by bringing the current operators into the Kubo formula), the positions of the resonant peaks can be readily figured out. For $\mathrm{Re}(\sigma_{zz})$, the positions of the resonant peaks do no change. However, for $\mathrm{Re}(\sigma_{xx})$, the main resonant peaks will undergo splitting and a position shift. Concretely, the interbranch $\omega = 2[\sqrt{n(n+1)}+\sqrt{n(n-1)}]\lambda eB$ peaks will be split into $\omega = \pm\omega_c + 2[\sqrt{(n+1)n}+\sqrt{n(n-1)}]\lambda eB$, and the intrabranch $\omega = 2[\sqrt{n(n+1)}-\sqrt{n(n-1)}]\lambda eB$ peaks will be split into $\omega = \pm\omega_c + 2[\sqrt{(n+1)n} - \sqrt{n(n-1)}]\lambda eB$.

More interestingly, as the degeneracy of the two chiral Landau levels is lifted by the symmetry-breaking term, an optical transition within the two chiral Landau levels is allowed. As the energy offset between the two chiral Landau levels is fixed to ω_c for arbitrary k_z, an absorption peak at $\omega = \omega_c$ is expected. Compared to the transitions between chiral Landau levels and nonchiral Landau levels, this transition is remarkable in the sense that it will not be influenced by the change in the chemical potential as long as the chiral Landau levels cross the Fermi level (if we restrict ourselves to the continuum model, this is always the case). Note that this effect is exclusive to multi-Weyl semimetals as there is no chiral Landau level degeneracy to be lifted for single Weyl semimetals.

Although we have chosen a simple diagonal term to discuss the effect of particle-hole symmetry breaking, the above conclusions will qualitatively hold for more complicated symmetry-breaking terms. From the results above, we can see that although the particle-hole symmetry breaking will affect the frequencies of the resonant peaks, the influence is, in fact, marginal and can be analyzed and regulated in real experiments. The linear scaling dependence on the magnetic field strength and the anisotropic behavior of the longitudinal conductivity still hold.

E. Generalization to higher monopole charge

The above discussions could be systematically extended to multi-Weyl semimetals with arbitrary monopole charge Q. Specifically, the number of chiral Landau levels equals Q, and the Landau levels behave as $E_n(0) \propto B^{Q/2}$, while the selection rules for optical conductivity remains unmodified.

Thus, for $Q \geqslant 3$, the scaling dependence of the resonant-peak frequencies on the magnetic field will follow the law $\omega \propto B^{Q/2}$, and the longitudinal conductivity will exhibit more anisotropic behaviors. This general observation can be utilized to determine the monopole charge of multi-Weyl semimetals.

IV. EXPERIMENTAL ESTIMATION

Now we give a brief experimental estimation. We take the material candidate $HgCr_2Se_4$ [28,29] as a concrete example. It was predicted that there are only two double Weyl points in $HgCr_2Se_4$ which are very close to the Fermi energy [28,29]; thus, it is an ideal platform to test the above predictions for the optical conductivity of double Weyl semimetals.

According to the band structure calculated in Ref. [29], we find $\lambda(\hbar\pi/a)^2 \approx 0.4$ eV with the lattice constant $a = 10.753$ Å [64], which gives $\lambda^{-1} \approx 1.63 m_e$. Then the interval between two near-neighbor resonant peaks is estimated to be $\Delta\omega \approx 4\lambda\hbar eB \approx 0.28$ meV for $B = 1$ T. To make the resonant peaks well separated in energy, the magnetic field can be chosen to be of the order of 10 T. In accordance with the magnetic field, adequate light for the magneto-optical experiment falls into the midinfrared regime. As the energy scale is of the order of several meV, the sample should be very clean, and the experimental temperature should be lower than 1 meV, i.e., $T < 10$ K.

V. DISCUSSIONS AND CONCLUSIONS

To conclude, in this paper, we studied the magneto-optical response of double Weyl semimetals. We found that the longitudinal components of the optical conductivity have a series of resonant peaks with their corresponding frequencies scaling linearly with the strength of the magnetic field and their frequency ratios following a special law determined by the underlying Landau level structure. Note that the linear scaling law is distinct from the characteristic square-root scaling law obeyed by the Dirac semimetals and single Weyl semimetals [61,62]. We also found that the optical conductivity is quite anisotropic, with two of the three longitudinal components residing in a linear background and the remaining one in a constant background. The anisotropy of the optical conductivity simply originates from the anisotropy of the energy dispersion of double Weyl semimetals. Because transitions related to the chiral Landau levels are strongly dependent on the positions of the Fermi energy, it was found that the tuning of the chemical potential duly modifies the corresponding resonant peaks of the optical conductivity.

Furthermore, we found that when the degeneracy of the chiral Landau levels is lifted by a particle-hole symmetry-breaking term, a new transition channel within the chiral Landau levels is opened for $\mathrm{Re}(\sigma_{xx})$, with the transition frequency directly controlled by the symmetry-breaking parameter and irresponsive to the change in chemical potential. The effect of finite temperature and impurity scattering on the optical conductivity is mainly a suppression of the resonant peaks, which indicates that our study is quite robust against these subtleties.

Among the above results, the linear scaling law between the frequencies of the resonant peaks and the magnetic field and the anisotropic behavior of the longitudinal conductivity are two most distinctive features which can be applied to determine whether the predicted candidate materials, like $HgCr_2Se_4$ [28,29] and $SrSi_2$ [30], are truly double Weyl semimetals or not.

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